

Neostigmine: Drug information - UpToDate

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For additional information see "Neostigmine: Patient drug information" and "Neostigmine: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Bloxiverz

Brand Names: Canada

- Neostigmine Omega

Pharmacologic Category

- Acetylcholinesterase Inhibitor

Dosing: Adult

Myasthenia gravis, treatment

Myasthenia gravis, treatment: *Canadian labeling:*

Oral: Initial: 15 mg administered 3 times daily. Usual daily dose: 150 mg/day in divided doses; interval between doses should be adjusted per patient response with larger doses provided at times of most fatigue. Usual daily dosage range: 15 to 375 mg/day.

Reversal of nondepolarizing neuromuscular blockade after surgery

Reversal of nondepolarizing neuromuscular blockade after surgery:

Note : An anticholinergic agent (atropine or glycopyrrolate) should be given intravenously prior to (if patient is bradycardic) or in conjunction with neostigmine (to prevent bradycardia and GI symptoms). Avoid use or use with caution in patients with certain neuromuscular diseases (eg, myasthenia gravis, muscular dystrophy); other agents may be preferred (eg, sugammadex); if neostigmine is used, reduced dosing may be required with titration based on train-of-four (TOF) monitoring (Ref). TOF monitoring should be used to determine time of neostigmine initiation and need for additional doses. Dosing may vary depending on patient specific factors (eg, depth of paralysis, type of anesthesia, organ dysfunction, age, acid-base status) (Ref). Refer to institutional protocols and policies.

Usual dose: IV: 0.02 to 0.07 mg/kg; generally achieves a TOF twitch ratio of 90% within 10 to 20 minutes of administration; maximum total dose: 0.07 mg/kg or 5 mg (whichever is less) (Ref).

Dose selection based on TOF count at end of surgery (Ref):

TOF count 1 or no TOF response: Delay reversal until TOF count ≥ 2 .

TOF count 2 or 3: 0.07 mg/kg; allow at least 15 to 30 minutes before tracheal extubation.

TOF count 4 with observable fade: 0.04 to 0.05 mg/kg; allow at least 10 to 15 minutes before tracheal extubation.

TOF count 4 with no observable fade: 0.02 mg/kg.

Acute colonic pseudo-obstruction

Acute colonic pseudo-obstruction (Ogilvie syndrome) (off-label use): **Note:** Continuous cardiac monitoring with clinical assessment for 30 minutes after administration. Atropine should be available at the bedside if bradyarrhythmia occurs (Ref).

Bolus: **IV:** 2 mg over 3 to 5 minutes (Ref).

Intermittent infusion: **IV:** 2.5 mg over 60 minutes (Ref).

Continuous infusion: **IV:** 0.4 mg per hour for up to 8 hours; if no response increase to 0.8 mg per hour for up to 24 hours (Ref).

Dosing: Kidney Impairment: Adult

There are no dosage adjustments provided in manufacturer's labeling.

The following adjustments have been recommended (Ref):

CrCl >50 mL/minute: No dosage adjustment necessary

CrCl 10 to 50 mL/minute: Administer 50% of normal dose.

CrCl <10 mL/minute: Administer 25% of normal dose.

Hemodialysis: No dosage adjustment necessary

Peritoneal dialysis: No dosage adjustment necessary

Continuous renal replacement therapy (CRRT): Administer 50% of normal dose

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in manufacturer's labeling; has not been studied.

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Neostigmine: Pediatric drug information")

Myasthenia gravis, juvenile

Myasthenia gravis, juvenile: Limited data available:

Diagnosis: **Note:** Perform in an area with adequate monitoring (eg, hospital setting); atropine should be available at bedside during administration (Ref).

Infants and Children <2 years: IM: 0.04 mg/kg once; if results equivocal or negative, may be repeated once in 4 hours (Ref).

Reversal of nondepolarizing neuromuscular blockade after surgery

Reversal of nondepolarizing neuromuscular blockade after surgery:

Note: An anticholinergic agent (atropine or glycopyrrolate) should be given prior to or in conjunction with neostigmine; in the presence of bradycardia, administer the anticholinergic prior to neostigmine. Peripheral nerve stimulation delivering train-of-four (TOF) stimulus must also be used to determine time of neostigmine initiation and need for additional doses. Refer to institutional protocol.

Infants, Children, and Adolescents:

Usual dose: IV: 0.03 to 0.07 mg/kg generally achieves a TOF twitch ratio of 90% within 10 to 20 minutes of administration; additional doses may be required; maximum total dose: 0.07 mg/kg or 5 mg (whichever is less)

Dose selection guide:

The 0.03 mg/kg dose is recommended for reversal of neuromuscular blocking agents with shorter half-lives (eg, rocuronium); or when the first twitch response to the TOF stimulus is substantially >10% of baseline or when a second twitch is present.

The 0.07 mg/kg dose is recommended for neuromuscular blocking agents with longer half-lives (eg, vecuronium, pancuronium); or when the first twitch response is relatively weak (ie, not substantially >10% of baseline); or rapid recovery is needed.

Dosing: Kidney Impairment: Pediatric

The manufacturer recommends close monitoring in patients with impaired kidney function. Based on adult information, dosage adjustment may be necessary.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Reported adverse reactions are for adults.

1% to 10%:

Cardiovascular: Atrioventricular block, cardiac arrhythmia (including atrioventricular nodal arrhythmia, bradycardia [Ruetzler 2022], tachycardia), flushing, hypotension, syncope

Dermatologic: Diaphoresis, pruritus, skin rash, urticaria

Gastrointestinal: Flatulence, increased peristalsis, nausea, sialorrhea, vomiting, xerostomia

Genitourinary: Urinary frequency

Hematologic & oncologic: Oxygen desaturation

Hypersensitivity: Anaphylaxis (Ruetzler 2022), hypersensitivity reaction

Nervous system: Asthenia, dizziness, drowsiness, dysarthria, headache, insomnia, loss of consciousness, postanesthetic shivering, seizure

Neuromuscular & skeletal: Arthralgia, muscle cramps, muscle spasm

Ophthalmic: Miosis, visual disturbance

Respiratory: Apnea, bronchospasm (Ruetzler 2022), dyspnea, increased bronchial secretions (including pharyngeal secretions), respiratory depression

Postmarketing:

Cardiovascular: ECG changes (nonspecific)

Gastrointestinal: Abdominal cramps, diarrhea

Neuromuscular & skeletal: Fasciculations

Contraindications

Hypersensitivity to neostigmine or any component of the formulation; peritonitis or mechanical obstruction of the intestinal or urinary tract.

Canadian labeling (tablets only): Additional contraindications (not in US labeling): Hypersensitivity to bromides.

Warnings/Precautions

Concerns related to adverse effects:

- Cardiovascular effects: Bradycardia, hypotension, and dysrhythmias may occur, particularly with IV use; risk may be increased in patients with certain cardiovascular conditions (eg, coronary artery disease, cardiac arrhythmias, recent acute coronary syndrome). Risk may also be increased in patients with myasthenia gravis. When IV neostigmine is administered for the reversal of nondepolarizing neuromuscular-blocking agents, atropine or glycopyrrolate should be administered concurrently or prior to neostigmine to lessen the risk of bradycardia.
- Cholinergic crisis: Overdosage may result in cholinergic crisis, characterized by extreme muscle weakness and potentially fatal respiratory paralysis. Cholinergic crisis should be distinguished from myasthenic crisis, which is also characterized by extreme muscle weakness, but would require radically different treatment.
- Hypersensitivity reactions: Symptoms of hypersensitivity have included anaphylaxis, angioedema, bradycardia, bronchospasm, erythema multiforme, facial swelling, flushing, generalized rash, hypotension, peripheral edema, pyrexia, and urticaria. Have atropine and epinephrine ready to treat hypersensitivity reactions.
- Neuromuscular effects: Large doses of IV neostigmine administered for the reversal of nondepolarizing neuromuscular-blocking agents when neuromuscular blockade is minimal can result in neuromuscular dysfunction. Reduce dose if recovery from neuromuscular blockade is nearly complete.

Disease-related concerns:

- Asthma: Use with caution in patients with asthma.
- Cardiovascular disease: Use with caution in patients with bradycardia, cardiac arrhythmias, coronary artery disease, or recent acute coronary syndrome.
- Hyperthyroidism: Use with caution in patients with hyperthyroidism.
- Megacolon/GI dysfunction: Large oral doses should be avoided with megacolon or decreased GI motility. Neostigmine may accumulate; toxicity may result when motility is restored.
- Neuromuscular diseases: For reversal of nondepolarizing neuromuscular blockade, avoid use or use with caution in patients with certain neuromuscular diseases (eg, myasthenia gravis, muscular dystrophy); other agents may be preferred (eg, sugammadex); if used, reduced dosing may be required with titration based on train-of-four monitoring (Balaka 2011; Briggs 2003; Buzello 1982). Adequate facilities should be available for cardiopulmonary resuscitation when testing and adjusting dose for myasthenia gravis.
- Peptic ulcer disease: Use with caution in patients with peptic ulcer disease.
- Seizure disorder: Use with caution in patients with epilepsy.
- Vagotonia: Use with caution in patients with vagotonia.

Special populations:

- Pediatric: When used IV for the reversal of nondepolarizing muscle relaxants, pediatric and adult dosing is similar. However, recovery may be more rapid and risk of complications may be greater in infants and small children. Monitor closely.
- Older adult: Use with caution and monitor for a longer period. Elderly patients experience slower spontaneous recovery from neuromuscular blocking agents.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Intravenous, as methylsulfate:

Bloxiverz: 5 mg/10 mL (10 mL [DSC]); 10 mg/10 mL (10 mL) [contains phenol]

Generic: 5 mg/10 mL (10 mL); 10 mg/10 mL (10 mL)

Solution Prefilled Syringe, Intravenous, as methysulfate:

Bloxiverz: 5 mg/5 mL (5 mL)

Generic: 3 mg/3 mL (3 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Bloxiverz Intravenous)

10 mg/10 mL (per mL): \$2.20

Solution (Neostigmine Methysulfate Intravenous)

5 mg/10 mL (per mL): \$0.25 - \$3.24

10 mg/10 mL (per mL): \$0.26 - \$3.36

Solution Prefilled Syringe (Bloxiverz Intravenous)

5 mg/5 mL (per mL): \$2.96

Solution Prefilled Syringe (Neostigmine Methysulfate Intravenous)

3 mg/3 mL (per mL): \$6.00

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection:

Generic: 2.5 mg/mL (5 mL)

Solution, Injection, as methysulfate:

Generic: 0.5 mg/mL (1 mL, 10 mL); 1 mg/mL (1 mL, 10 mL)

Administration: Adult

Tablets (neostigmine bromide): Prostigmin [Canadian product]: Muscarinic effects may be decreased when administered with food or milk. Consider giving larger portions of the daily dose around fatigue prone times (eg, mealtimes, afternoons).

Injectable (neostigmine methysulfate):

Neostigmine Omega, Prostigmin [Canadian products]: May be administered IM, IV, or SUBQ.

Bloxiverz: Administer by slow IV injection over at least 1 minute.

Acute colonic pseudo-obstruction (off-label use):

Bolus: IV: Administer over 3 to 5 minutes (Ref).

Intermittent infusion: IV: Administration over 60 minutes (Ref).

Continuous infusion: IV: Refer to indication-specific infusion rates in "Dosing: Adult" for detailed recommendations (Ref).

Preparation for Administration: Adult

Acute colonic pseudo-obstruction (off-label use):

Intermittent infusion: Dilute 2.5 mg in 100 mL normal saline to a final concentration of 0.025 mg/mL (Ref).

Continuous infusion: Dilute 5 mg in 50 mL normal saline to a final concentration of 0.1 mg/mL. Refer to indication-specific infusion rates in "Dosing: Adult" for detailed recommendations (Ref).

Administration: Pediatric

Parenteral: Route of administration dependent upon indication and age; use caution.

IV: Reversal of nondepolarizing neuromuscular blockade after surgery: Neonates, Infants, Children, and Adolescents: Administer undiluted by slow IV injection over several minutes.

IM: Myasthenia gravis, diagnosis and treatment: Neonates, Infants, Children, and Adolescents: May administer undiluted IM.

SUBQ: Myasthenia gravis, treatment: Neonates: May administer undiluted SUBQ.

Storage/Stability

Injection:

Bloxiverz: Store between 20°C and 25°C (68°F and 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F). Protect from light. Store in carton until time of use.

Neostigmine Omega [Canadian product]: Store at 15°C to 30°C; protect from light; multiple use vials should be discarded 28 days after initial use.

Prostigmin [Canadian product]: Store at 15°C to 30°C; protect from light.

Tablets: Prostigmin [Canadian product]: Store at 15°C to 30°C; protect from light.

Use: Labeled Indications

US labeling:

Reversal of nondepolarizing muscle relaxants: Reversal of effects of nondepolarizing neuromuscular blocking agents after surgery.

Canadian labeling:

Myasthenia gravis: Symptomatic control of myasthenia gravis (oral product).

Limitations of use: Other agents may be preferred for symptomatic treatment or acute exacerbations.

Use: Off-Label: Adult

Acute colonic pseudo-obstruction (Ogilvie's syndrome)

Medication Safety Issues

Sound-alike/look-alike issues:

Bloxiverz may be confused with Vazculep (phenylephrine injection) due to similar packaging

Prostigmin may be confused with physostigmine

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Agents with Clinically Relevant Anticholinergic Effects: Acetylcholinesterase Inhibitors may decrease therapeutic effects of Agents with Clinically Relevant Anticholinergic Effects. Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Acetylcholinesterase Inhibitors. *Risk C: Monitor*

Amifampridine: Acetylcholinesterase Inhibitors may increase therapeutic effects of Amifampridine. Amifampridine side effects may also be increased. Amifampridine may increase therapeutic effects of Acetylcholinesterase Inhibitors. Acetylcholinesterase inhibitor side effects may also be increased. *Risk C: Monitor*

Benoxinate: Acetylcholinesterase Inhibitors may increase therapeutic effects of Benoxinate. Specifically, the effects of benoxinate may be prolonged. *Risk C: Monitor*

Beta-Blockers: Acetylcholinesterase Inhibitors may increase bradycardic effects of Beta-Blockers. *Risk C: Monitor*

Bradycardia-Causing Agents: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Ceritinib: Bradycardia-Causing Agents may increase bradycardic effects of Ceritinib. Management: If this combination cannot be avoided, monitor patients for evidence of symptomatic bradycardia, and closely monitor blood pressure and heart rate during therapy. *Risk D: Consider Therapy Modification*

Chlorprothixene: Acetylcholinesterase Inhibitors may increase adverse/toxic effects of Chlorprothixene. Acetylcholinesterase Inhibitors may increase therapeutic effects of Chlorprothixene. *Risk C: Monitor*

Cholinergic Agonists: Acetylcholinesterase Inhibitors may increase adverse/toxic effects of Cholinergic Agonists. Specifically, cholinergic effects may be enhanced or increased. *Risk C: Monitor*

Corticosteroids (Systemic): May increase adverse/toxic effects of Acetylcholinesterase Inhibitors. Increased muscular weakness may occur. *Risk C: Monitor*

Dipyridamole: May decrease therapeutic effects of Acetylcholinesterase Inhibitors. *Risk C: Monitor*

Etrasimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Fexinidazole: Bradycardia-Causing Agents may increase arrhythmogenic effects of Fexinidazole. *Risk X: Avoid*

Fingolimod: Bradycardia-Causing Agents may increase bradycardic effects of Fingolimod. Management: Consult with the prescriber of any bradycardia-causing agent to see if the agent could be switched to an agent that does not cause bradycardia prior to initiating fingolimod. If combined, perform continuous ECG monitoring after the first fingolimod dose. *Risk D: Consider Therapy Modification*

Gepotidacin: May increase therapeutic effects of Acetylcholinesterase Inhibitors. *Risk C: Monitor*

Ivabradine: Bradycardia-Causing Agents may increase bradycardic effects of Ivabradine. *Risk C: Monitor*

Kanamycin: Neostigmine (Systemic) may decrease therapeutic effects of Kanamycin. *Risk C: Monitor*

Lacosamide: Bradycardia-Causing Agents may increase AV-blocking effects of Lacosamide. *Risk C: Monitor*

Landiolol: Bradycardia-Causing Agents may increase bradycardic effects of Landiolol. *Risk X: Avoid*

Midodrine: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Neuromuscular-Blocking Agents (Nondepolarizing): Acetylcholinesterase Inhibitors may decrease neuromuscular-blocking effects of Neuromuscular-Blocking Agents (Nondepolarizing). *Risk C: Monitor*

Ozanimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Ponesimod: Bradycardia-Causing Agents may increase bradycardic effects of Ponesimod. Management: Avoid coadministration of ponesimod with drugs that may cause bradycardia when possible. If combined, monitor heart rate closely and consider obtaining a cardiology consult. Do not initiate ponesimod in patients on beta-blockers if HR is less than 55 bpm. *Risk D: Consider Therapy Modification*

Siponimod: Bradycardia-Causing Agents may increase bradycardic effects of Siponimod. Management: Avoid coadministration of siponimod with drugs that may cause bradycardia. If combined, consider obtaining a cardiology consult regarding patient monitoring. *Risk D: Consider Therapy Modification*

Succinylcholine: Acetylcholinesterase Inhibitors may increase neuromuscular-blocking effects of Succinylcholine. *Risk C: Monitor*

Pregnancy Considerations

Adverse events were not observed in animal reproduction studies (doses used were below maximum expected human exposure based on BSA).

When used as adjunct to analgesia in labor, adverse events to the fetus and mother are dose- and route-dependent (Habib 2006).

Neostigmine may be used to treat myasthenia gravis in pregnant patients; however, if an acetylcholinesterase inhibitor is needed during pregnancy, another agent may be preferred (Sanders 2016).

Breastfeeding Considerations

It is not known if neostigmine is present in breast milk.

According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother.

Monitoring Parameters

ECG, blood pressure, and heart rate especially with IV use; train-of-four monitoring; consult individual institutional policies and procedures.

Acute colonic pseudo-obstruction (off-label use): Keep patient supine upon administration. Patient should receive continuous ECG monitoring with clinical assessment for 30 minutes after administration (Saunders 2005).

Mechanism of Action

Inhibits destruction of acetylcholine by acetylcholinesterase which facilitates transmission of impulses across myoneural junction; direct cholinomimetic effect on skeletal muscle and possible on autonomic ganglion cells and neurons of the CNS

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Peristaltic activity: Oral: 2 to 4 hours; Parenteral: 10 to 30 minutes

Duration: IM: 2.5 to 4 hours

Absorption: Oral: Poor (~1% to 2%)

Distribution: V_d : IV: 0.12 to 1.4 L/kg

Protein binding: 15% to 25% to albumin

Metabolism: Hepatic

Half-life elimination:

IM: Adults: 51 to 90 minutes

IV: Range: 24 to 113 minutes

Infants 2 to 10 months: Mean: 39 ± 5 minutes

Children 1 to 6 years: Mean: 48 ± 16 minutes

Adults 29 to 48 years: 67 ± 8 minutes

Anephric patients: 181 ± 54 minutes

Renal transplant recipients: 104.7 ± 64 minutes

Oral: Adults: 42 to 60 minutes

Time to peak: Oral: 1 to 2 hours (Aquilonius 1986)

Excretion: Urine (50% as unchanged drug; remainder as metabolites) (Aquilonius 1986)

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Altered kidney function: Elimination half-life prolonged in anephric patients.

Patient Counseling Points

(For additional information see "Neostigmine: Patient drug information")

What is this drug used for?

- It is used to treat myasthenia gravis.
- It is used to get back function in muscles after surgery.
- It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Dizziness or headache
- Upset stomach or throwing up
- Dry mouth
- Trouble sleeping

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

All products:

- Muscle weakness
- Twitching
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Injection:

- Swelling in the arms or legs
- Fever
- Flushing

- Shivering
- Chest pain
- Fast, slow, or abnormal heartbeat
- Shortness of breath
- Very bad dizziness or passing out
- Seizures
- Feeling very sleepy
- Change in eyesight

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Prostigmin;
- (AR) Argentina: Neostigmina metilsulfato larjan | Prostigmin;
- (AT) Austria: Normastigmin | Prostigmin;
- (BD) Bangladesh: Prostigmin;
- (BE) Belgium: Prostigmine;
- (BG) Bulgaria: Neostigmin | Syntostigmin;
- (BR) Brazil: Prostigmine;
- (CL) Chile: Neostigmina Bromuro | Prostigmine;
- (CN) China: Neostigmine br.;
- (CO) Colombia: Prostigmine;
- (CR) Costa Rica: Neostigmina Bromuro;
- (CZ) Czech Republic: Neoeserin | Prostigmin | Syntostigmin;
- (DO) Dominican Republic: Intrastigmina | Prostigmine;
- (EC) Ecuador: Prostigmine;
- (EE) Estonia: Neostigmin | Neostigmine methylsulfate | Proserin | Prostigmin | Syntostigmin;
- (EG) Egypt: Amostigmine | Neostigmine | Prostigmine;
- (ES) Spain: Prostigmine;
- (FI) Finland: Metastigmin | Neostigmin | Neostigmin fortior;
- (FR) France: Neostigmine Opi;
- (GB) United Kingdom: Neostigmine;
- (GR) Greece: Neostigmina lph | Neostigmine | Prostigmine;
- (HU) Hungary: Prostigmin | Stigmosan;
- (ID) Indonesia: Neostigmine | Prostigmin | Tyzox;
- (IE) Ireland: Neostigmine;

- (IL) Israel: Prostigmine;
- (IN) India: Tilstigmin;
- (IT) Italy: Intrastigmina | Prostigmina;
- (JO) Jordan: Prostigmine;
- (KR) Korea, Republic of: Neostigmin;
- (KW) Kuwait: Prostigmine;
- (LB) Lebanon: Prostigmine;
- (LT) Lithuania: Neostigmin | Neostigmina lph | Proserin | Prostigmin | Syntostigmin;
- (LU) Luxembourg: Prostigmine;
- (LV) Latvia: Neostigmin | Proserin | Prostigmin | Syntostigmin;
- (MA) Morocco: Prostigmine;
- (MX) Mexico: Neostigmina | Prostigmine;
- (MY) Malaysia: Neomin | Neostigmine;
- (NL) Netherlands: Prostigmin;
- (PE) Peru: Prostigmine;
- (PH) Philippines: Prostigmine;
- (PK) Pakistan: Neostigmine | Prostigmin;
- (PL) Poland: Polstigminum | Prostigmin | Syntostigmin;
- (PR) Puerto Rico: Bloxiverz | Neostigmine methysulfate | Prostigmin;
- (PT) Portugal: Prostigmine;
- (QA) Qatar: Epistigmin | Prostigmin;
- (RU) Russian Federation: Proserin;
- (SA) Saudi Arabia: Neostigmine | Prostigmine;
- (SE) Sweden: Neostigmin;
- (SG) Singapore: Neostigmine;
- (SI) Slovenia: Prostigmin;
- (SK) Slovakia: Prostigmin | Syntostigmin;
- (TN) Tunisia: Prostigmine;
- (TR) Turkey: Neostigmin | Prostigmine;
- (TW) Taiwan: Neostigmin | Neostigmine | Neostimin | Prostigmin;
- (UA) Ukraine: Proserin;
- (UY) Uruguay: Prostigmine;
- (VE) Venezuela, Bolivarian Republic of: Prostigmine

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REFERENCES

1. Abeyta BJ, Albrecht RM, Schermer CR. Retrospective study of neostigmine for the treatment of acute colonic pseudo-obstruction. *Am Surg*. 2001;67(3):265-269. [PubMed 11270887]
2. Amaro R, Rogers AI. Neostigmine infusion: new standard of care for acute colonic pseudo-obstruction? *Am J Gastroenterol*. 2000;95(1):304-305. doi:10.1111/j.1572-0241.2000.01737.x [PubMed 10638606]
3. Aquilonius SM, Hartviq P. Clinical pharmacokinetics of cholinesterase inhibitors. *Clin Pharmacokinet*. 1986;11(3):236-249. [PubMed 3524957]
4. Aronoff GR, Bennett WM, Berns JS, et al. *Drug Prescribing in Renal Failure: Dosing Guidelines for Adults and Children, 5th ed*. Philadelphia, PA: American College of Physicians; 2007.
5. Balaka C, Poulida S, Miliatou M, et al. Comparison of sugammadex to neostigmine reversal of neuromuscular blockade in patients with myasthenia gravis. *Journal of Cardiothoracic and Vascular Anesthesia*. 2011;25:S22-S23.
6. Bloxiverz (neostigmine) [prescribing information]. Lenoir, NC: Exela Pharma Sciences LLC; December 2023.

7. Bricoune O, Hamner B, Gieron-Korthals M. Challenges in diagnosing and treating myasthenia gravis in infants and children with presentation of cases. *Adv Pediatr.* 2023;70(1):81-90. doi:10.1016/j.yapd.2023.03.007 [PubMed 37422299]
8. Briggs ED, Kirsch JR. Anesthetic implications of neuromuscular disease. *J Anesth.* 2003;17(3):177-185. doi:10.1007/s00540-003-0169-5 [PubMed 12911205]
9. Buzello W, Krieg N, Schlickewei A. Hazards of neostigmine in patients with neuromuscular disorders. Report of two cases. *Br J Anaesth.* 1982;54(5):529-534. doi:10.1093/bja/54.5.529 [PubMed 7073921]
10. Donaldson JO, Penn AS, Lisak RP, Abramsky O, Brenner T, Schotland DL. Antiacetylcholine receptor antibody in neonatal myasthenia gravis. *Am J Dis Child.* 1981;135(3):222-226. doi:10.1001/archpedi.1981.02130270014006 [PubMed 6163352]
11. Gropper M. Qualitative monitoring used (peripheral nerve stimulator). *Miller's Anaesth.* 2019;1(9):3170.
12. Habib AS and Gan TJ, "Use of Neostigmine in the Management of Acute Postoperative Pain and Labour Pain: A Review," *CNS Drugs*, 2006, 20(10):821-39. [PubMed 16999453]
13. Hassoun M, El Turjuman U, Chokr I, Fakhoury H. Myasthenia gravis in the neonate. *NeoReviews.* 2010;11(4):200-205.
14. İlban Ö, Çiçekçi F, Çelik JB, Baş MA, Duman A. Neostigmine treatment protocols applied in acute colonic pseudo-obstruction disease: a retrospective comparative study. *Turk J Gastroenterol.* 2019;30(3):228-233. doi:10.5152/tjg.2018.18193 [PubMed 30541715]
15. Lindroos JLV, Bjørk MH, Gilhus NE. Transient neonatal myasthenia gravis as a common complication of a rare disease: a systematic review. *J Clin Med.* 2024;13(4):1136. doi:10.3390/jcm13041136 [PubMed 38398450]
16. Manzur AY. Disorders of neuromuscular transmission and of motor neurons. In: Kliegman RM, St. Geme J, Blum NJ, eds. *Nelson Textbook of Pediatrics*. 22nd ed. Elsevier; 2024:chap. 652.
17. Neostigmine Omega (neostigmine) [product monograph]. Montreal, Quebec, Canada: Omega Laboratories Limited; May 2012.
18. Neostigmine injection [prescribing information]. Eatontown, NJ: West Ward Pharmaceuticals; October 2011.
19. Neostigmine injection [prescribing information]. Princeton, NJ: Dr. Reddy's Laboratories Inc; February 2021.
20. Neostigmine Methylsulfate Injection [product monograph]. Boucherville, Quebec, Canada: JAMP Pharma Corporation; April 2023.
21. Neostigmine Methylsulfate Injection [prescribing information]. Lake Zurich, IL: Fresenius Kabi; March 2020.
22. Neostigmine Methylsulfate Injection [prescribing information]. Lake Zurich, IL: Fresenius Kabi; January 2024.
23. Papazian O. Transient neonatal myasthenia gravis. *J Child Neurol.* 1992;7(2):135-141. doi:10.1177/088307389200700202 [PubMed 1573230]
24. Paran H, Silverberg D, Mayo A, Shwartz I, Neufeld D, Freund U. Treatment of acute colonic pseudo-obstruction with neostigmine. *J Am Coll Surg.* 2000;190(3):315-318. [PubMed 10703857]
25. Ponc R, Saunders MD, and Kimmey MB, "Neostigmine for the Treatment of Acute Colonic Pseudo-Obstruction," *N Engl J Med*, 1999, 341(3):137-41. [PubMed 10403850]
26. Prostigmin (neostigmine) [product monograph]. Montreal, Quebec, Canada: Valeant Canada; June 2008.
27. Refer to manufacturer's labeling.

28. Ruetzler K, Li K, Chhabada S, et al. Sugammadex versus neostigmine for reversal of residual neuromuscular blocks after surgery: a retrospective cohort analysis of postoperative side effects. *Anesth Analg*. 2022;134(5):1043-1053. doi:10.1213/ANE.0000000000005842 [PubMed 35020636]
29. Sanders DB, Wolfe GI, Benatar M, et al. International consensus guidance for management of myasthenia gravis: executive summary. *Neurology*. 2016;87(4):419-425. doi:10.1212/WNL.0000000000002790 [PubMed 27358333]
30. Santiago Gonçalves C, Nunes Marques MI, Antunes S, Serrano A. Transient neonatal myasthenia gravis: a case report. *Cureus*. 2021;13(12):e20592. doi:10.7759/cureus.20592 [PubMed 35103168]
31. Saunders MD, Kimmey MB. Systematic review: acute colonic pseudo-obstruction. *Aliment Pharmacol Ther*. 2005;22(10):917-925. [PubMed 16268965]
32. Silvestri NJ, Wolfe GI. Myasthenia gravis. *Semin Neurol*. 2012;32(3):215-226. [PubMed 23117946]
33. Trevisani GT, Hyman NH, Church JM. Neostigmine: safe and effective treatment for acute colonic pseudo-obstruction. *Dis Colon Rectum*. 2000;43(5):599-603. [PubMed 10826417]
34. Turégano-Fuentes F, Muñoz-Jiménez F, Del Valle-Hernández E, et al. Early resolution of Ogilvie's syndrome with intravenous neostigmine: a simple, effective treatment. *Dis Colon Rectum*. 1997;40(11):1353-1357. doi:10.1007/BF02050822 [PubMed 9369112]
35. Valle RG, Godoy FL. Neostigmine for acute colonic pseudo-obstruction: a meta-analysis. *Ann Med Surg (Lond)*. 2014;3(3):60-64. doi:10.1016/j.amsu.2014.04.002 [PubMed 25568788]

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